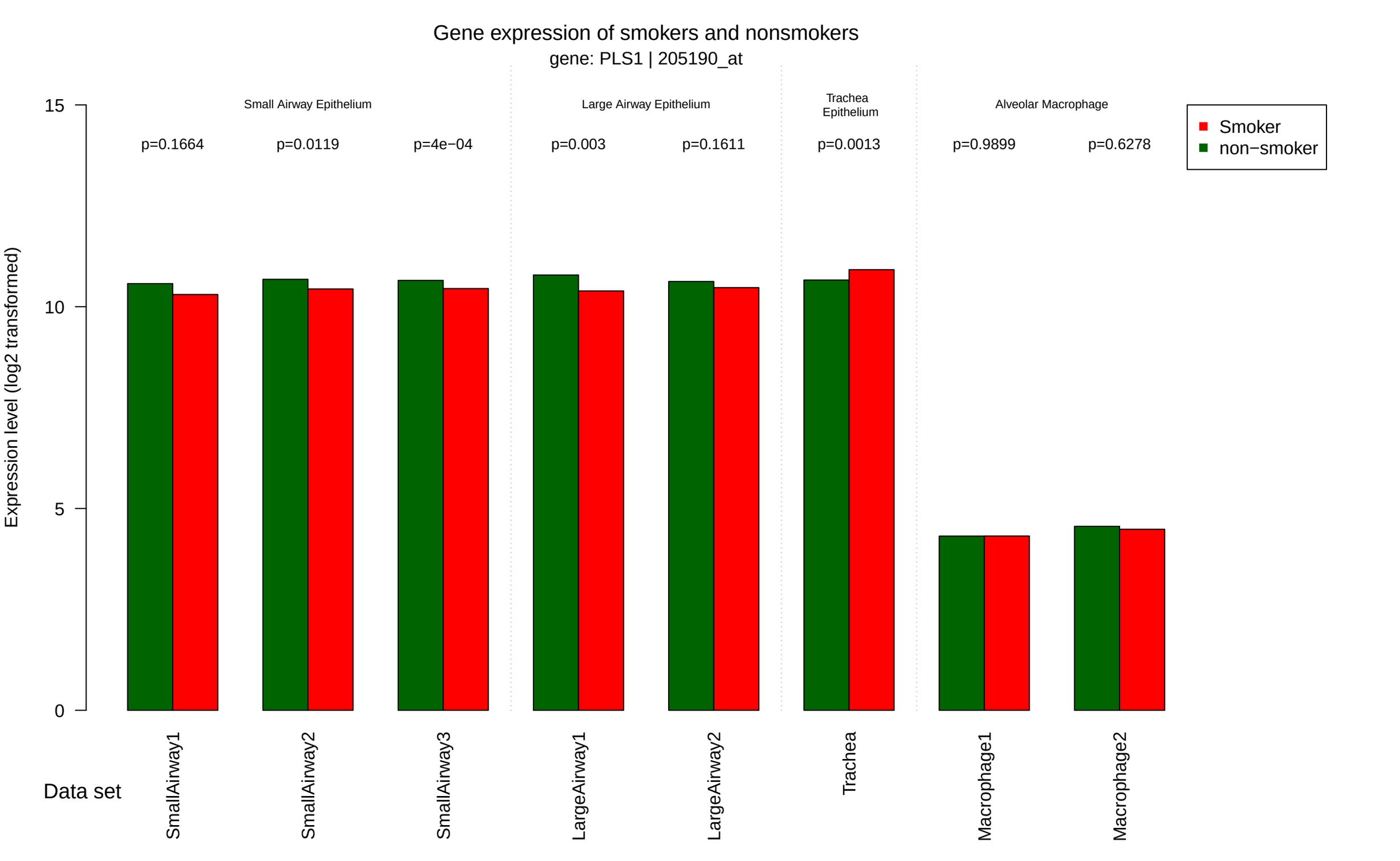
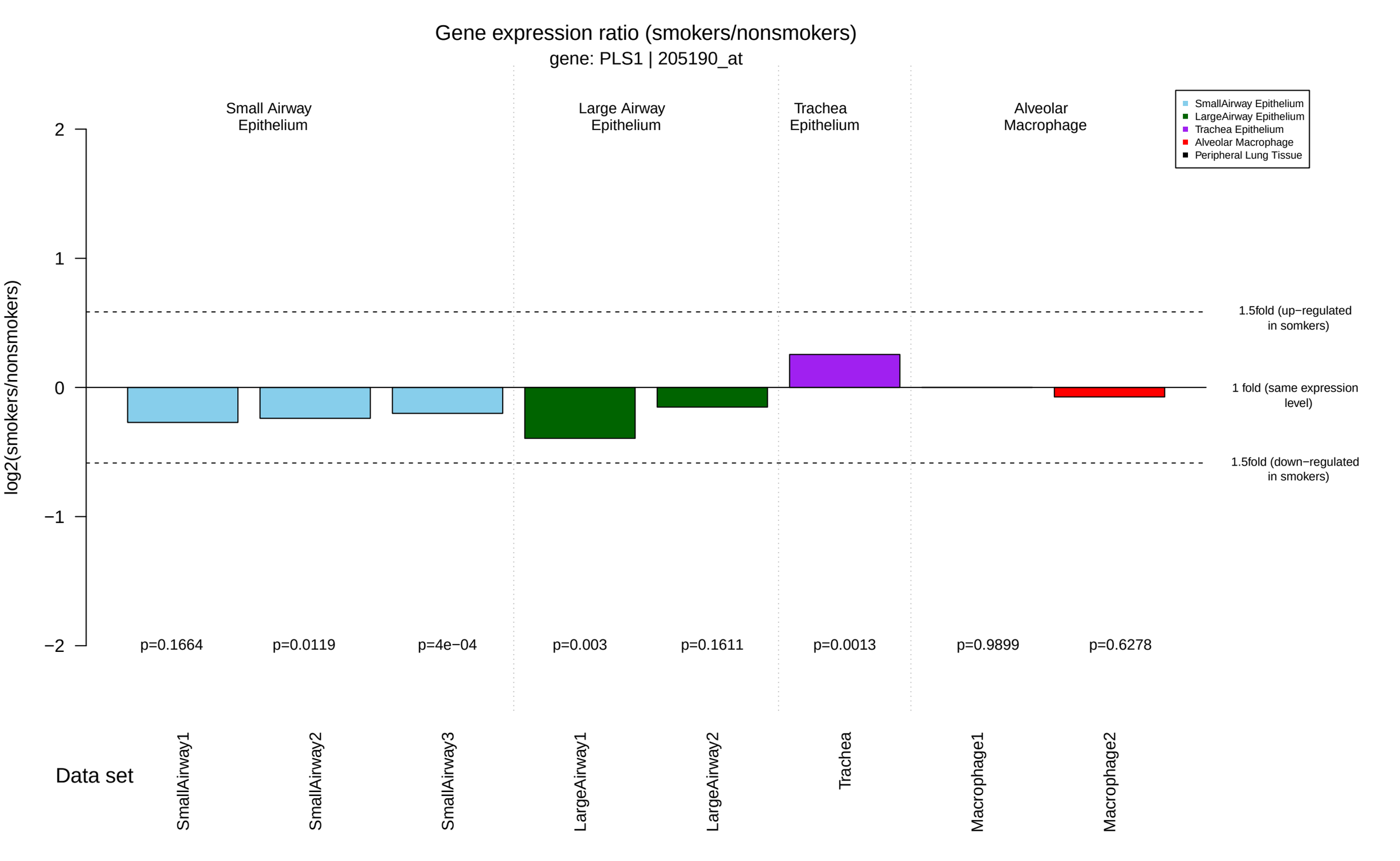
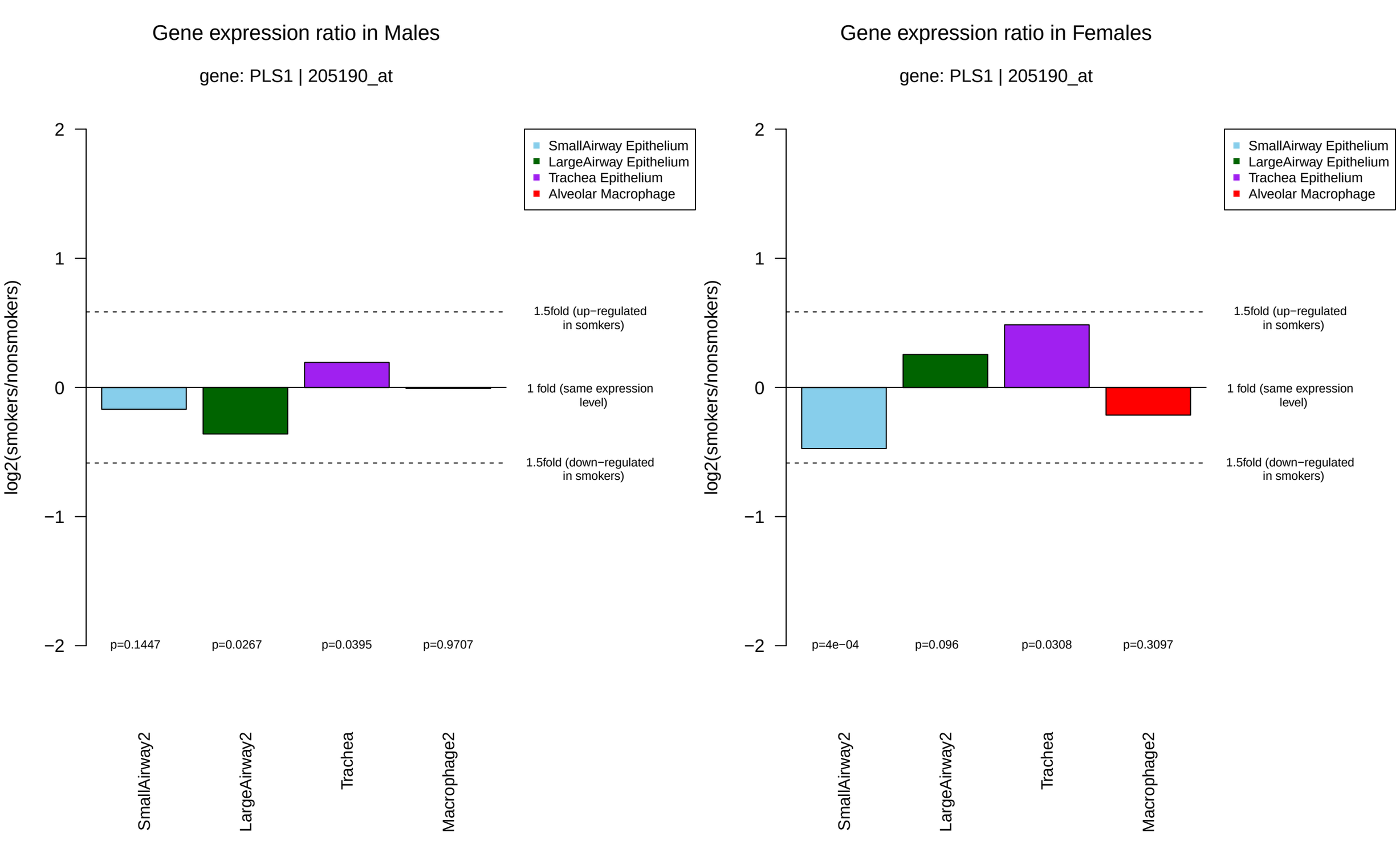
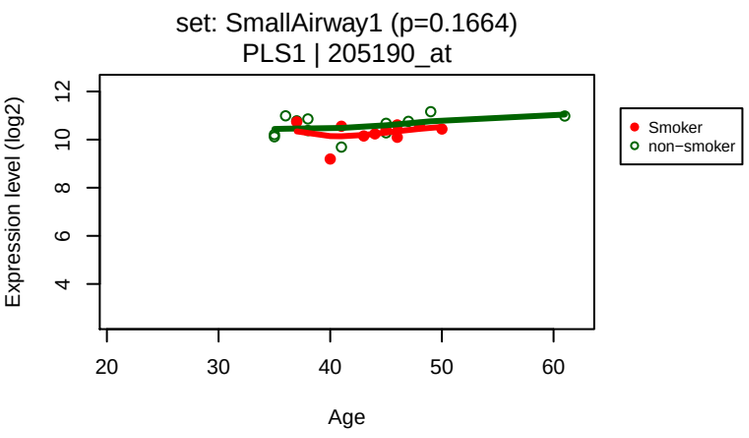
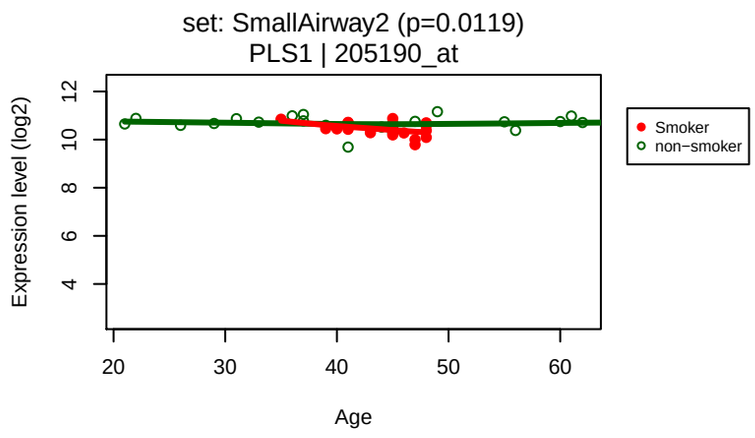
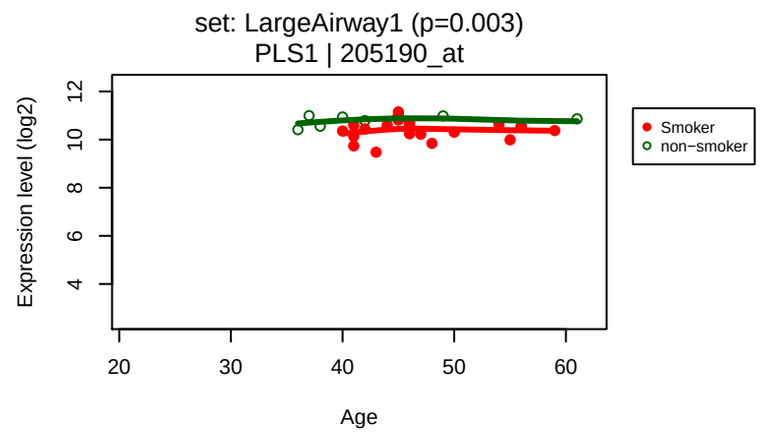

Gene expression of smokers and nonsmokers
gene: PLS1 | 205190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1664
p=0.0119
p=4e−04
p=0.003
p=0.1611
p=0.0013
p=0.9899
p=0.6278
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

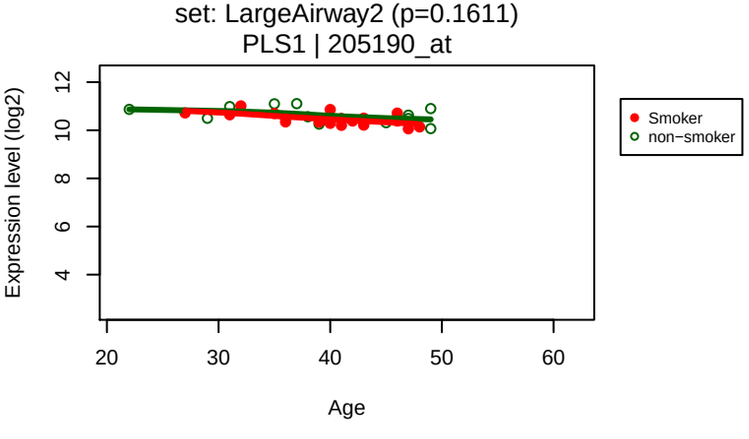

Gene expression ratio (smokers/nonsmokers)
gene: PLS1 | 205190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1664
p=0.0119
p=4e−04
p=0.003
p=0.1611
p=0.0013
p=0.9899
p=0.6278
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

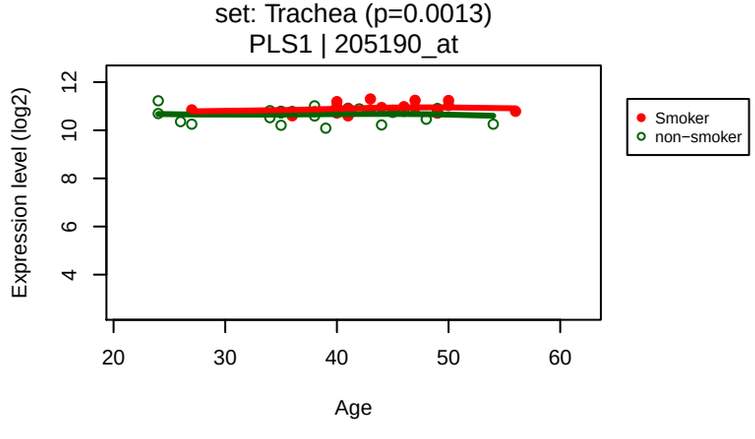

Gene expression ratio in Males
gene: PLS1 | 205190_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1447
p=0.0267
p=0.0395
p=0.9707
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLS1 | 205190_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=4e−04
p=0.096
p=0.0308
p=0.3097
SmallAirway2
LargeAirway2
Trachea
Macrophage2

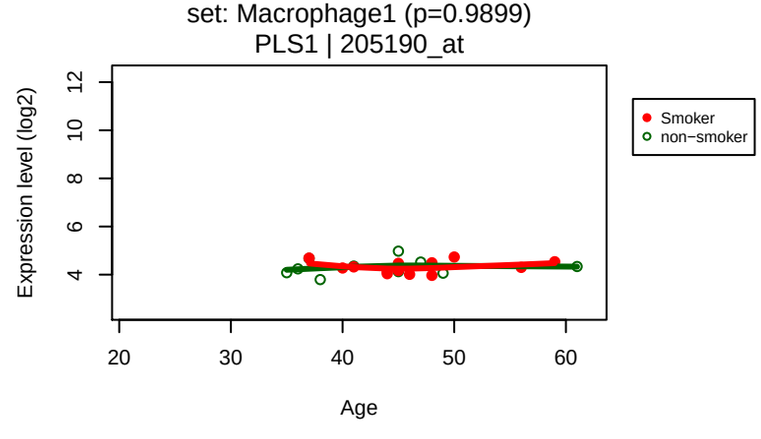

set: SmallAirway1 (p=0.1664)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker

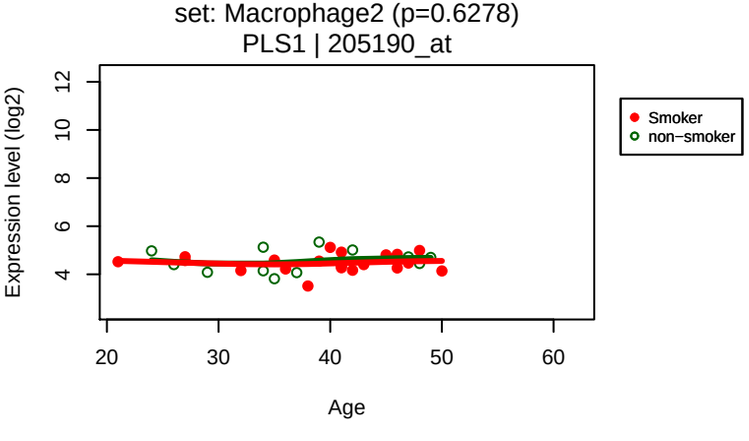

set: SmallAirway2 (p=0.0119)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker

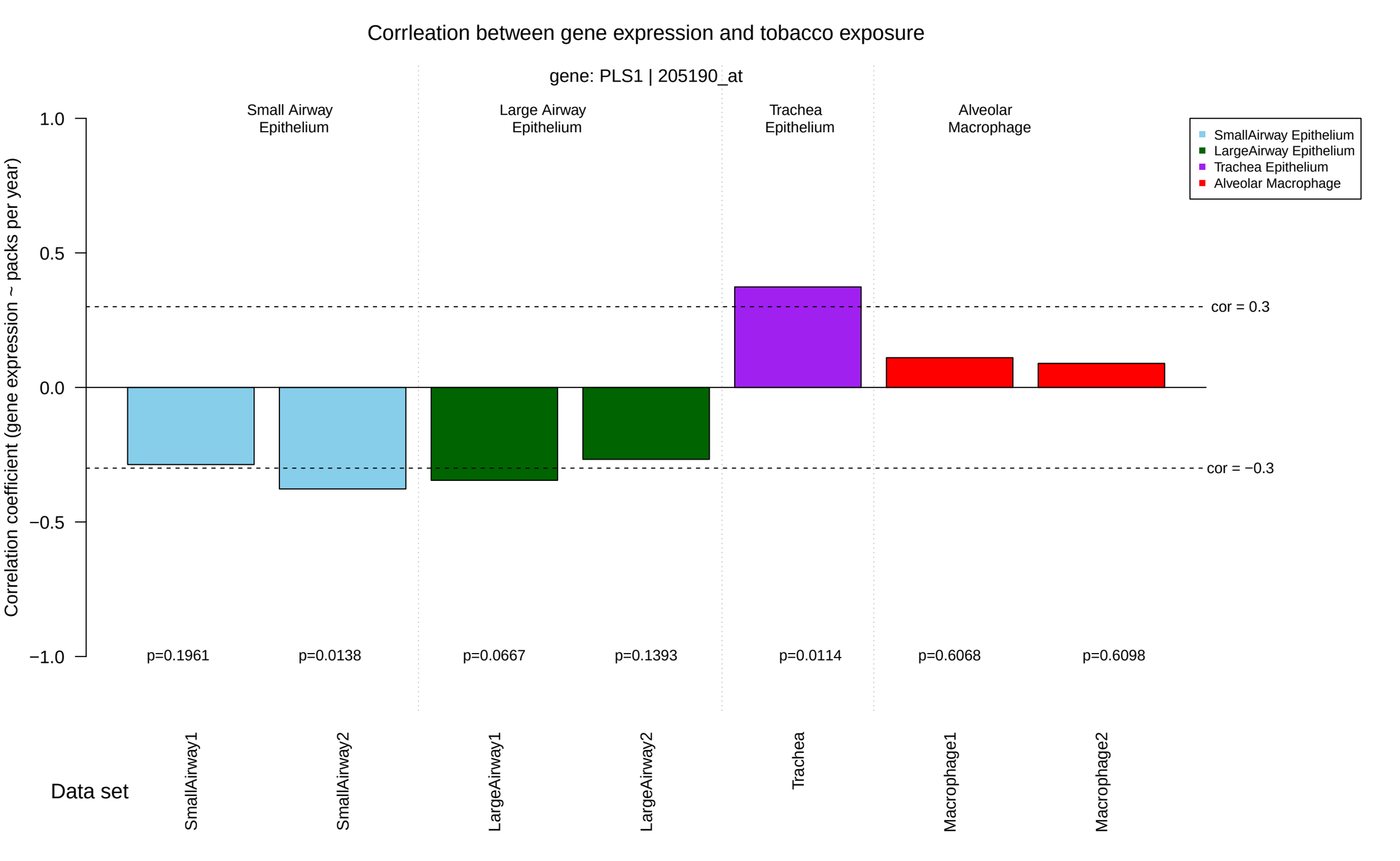

set: LargeAirway1 (p=0.003)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1611)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0013)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9899)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6278)
PLS1 | 205190_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLS1 | 205190_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1961
p=0.0138
p=0.0667
p=0.1393
p=0.0114
p=0.6068
p=0.6098
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2